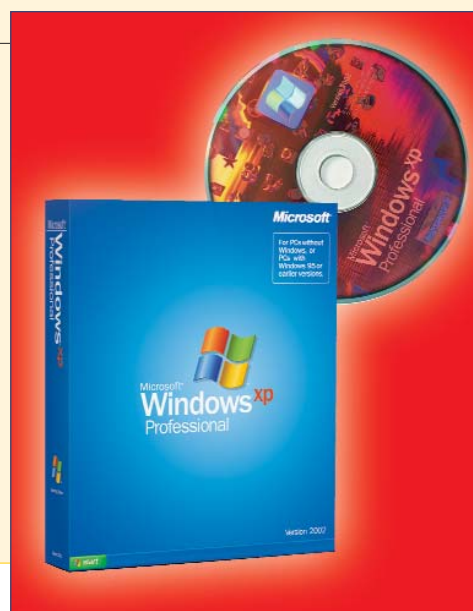


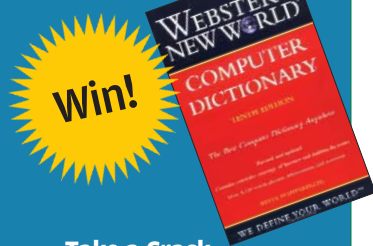
So you downloaded Service Pack 2 for Windows XP? How do you go about creating a bootable Windows XP installation CD with SP2 integrated but using your existing CD? Send your solutions to takeacrack@thinkdigit.com

THIS MONTH'S CHALLENGE

Creating a bootable Windows XP installation CD with SP2



LAST MONTH'S CHALLENGE



Take a Crack and Win

Webster's New World Computer Dictionary
By Bryan Pfaffenberger
Published by Pearson Education

Rules and Regulations

Readers are requested to send in their answers by the 15th of the month of publication.

Employees of Jasubhai Digital Media and their relatives are not permitted to participate in this contest.

Readers are encouraged to send their replies by e-mail. Jasubhai Digital Media will not entertain any unsolicited communication.

Jasubhai Digital Media is not responsible for any damage to your system that may be caused while you are solving the problem.

How To Safely Overclock Your CPU

Last month, we asked you how you could safely overclock your CPU. Unfortunately, none of the entries we received were good enough to deserve the prize. So here's the solution:

Most processors allow overclocking only to a certain extent. Others, by virtue of their design, allow it a little more. Most processors today have their clock multipliers locked and this prevents them from being tweaked through that component. In such cases, the bus speed needs to be used, but you need to take care when you do this so that the rest of the system is not adversely affected.

Keep in mind that your warranties will not cover any damage caused by overclocking; so be very sure of the extent to which you can push your system safely.

The first by-product of a tweaking experiment will be the generation of heat in your critical components. So before you embark on this task, ensure you have adequate cooling for components like the processor, RAM and motherboard chipset.

Steps To Overlock Your CPU

Consult the manual to make sure that your motherboard supports overclocking through

jumper settings or BIOS tweaks. Only when you are sure that you can change the CPU's core voltage, clock multiplier or bus speed, should you go about pushing your system harder. To overclock:

Enter the BIOS by rebooting your computer and hitting the [DEL] or the [F2] key. The exact steps will vary according to your BIOS type and version. Refer to the motherboard manual to help locate the FSB frequency and the CPU voltage settings.

Increase the CPU FSB frequency in small increments of, say, 2 MHz. Change the CPU Speed/Voltage Settings to 'Manual' to let you change them. Note: Throughout this exercise, monitor your system temperatures and voltages very closely. It is critical that they remain within limits for your system to run reliably. Failing to do so will result in shortening the life of your processor or even failure. It is important to start out your

overclocking at the rated settings and work your way up in small increments. Save your settings and reboot the machine. The PC should boot normally. Check for any signs of instability. Run a CPU-intensive program or game like Half-Life 2 to stress the CPU and make sure it is working fine. Reboot and enter the BIOS again. Increase the CPU voltage, again in small increments. Restart and repeat the previous step. Continue till you feel the machine has been pushed to its limits. Reboot and enter the BIOS again. Increase the CPU voltage, again in small increments. Restart and repeat the step 4. Continue the process till you feel the

Config System Frequency/Voltage		Up/Down
CPU FSB Frequency	1.750 V	
CPU Speed/Voltage Setting	1.725 V	
CPU Speed	1.700 V	
CPU Voltage	1.675 V	
DDR Voltage	1.650 V	
AGP Voltage	1.625 V	
V-Link Voltage	1.600 V	
Performance Mode	1.575 V	
	1.550 V	
	1.525 V	

Changing the CPU Voltage

machine has been pushed to its limits. A stable overclocked system is one that has the right balance between CPU FSB, voltage and bus speed settings. Overclocking is achieved by trial and error and we cannot overemphasise the importance of safety.

Config System Frequency/Voltage		Up/Down
CPU FSB Frequency	200 MHz	
CPU Speed/Voltage Setting	201 MHz	
CPU Speed	202 MHz	
CPU Voltage	203 MHz	
DDR Voltage	204 MHz	
AGP Voltage	205 MHz	
V-Link Voltage	206 MHz	
Performance Mode	207 MHz	

Changing the CPU FSB frequency